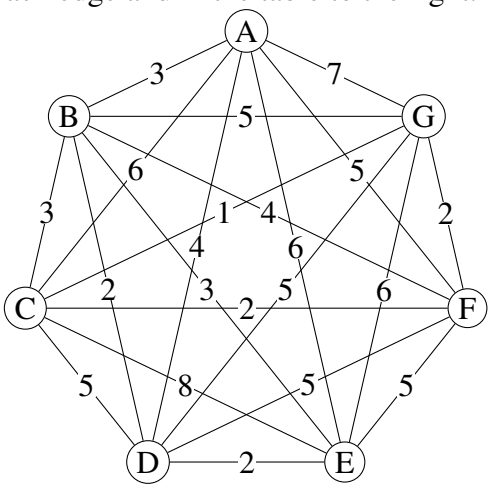
Name: _____ score:_____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

Questions 1 and 2 will use the weighted graph below. Edges weights are written in the middle of each edge and in the table to the right.



	A	B	C	D	E	F	G
A	0	3	6	4	6	5	7
B	3	0	3	2	3	4	5
C	6	3	0	5	8	2	1
D	4	2	5	0	2	5	5
E	6	3	8	2	0	5	6
F	5	4	2	5	5	0	2
G	7	5	1	5	6	2	0

1. (4 points) Use the Nearest Neighbor Algorithm (NNA) starting at vertex *G* to find a Hamiltonian circuit in the graph. Draw the circuit on the vertices below and determine weight of the circuit.

   Nearest Neighbor Algorithm: From the present vertex, move to the nearest **unvisited** vertex until all vertices are visited. Then close the circuit.

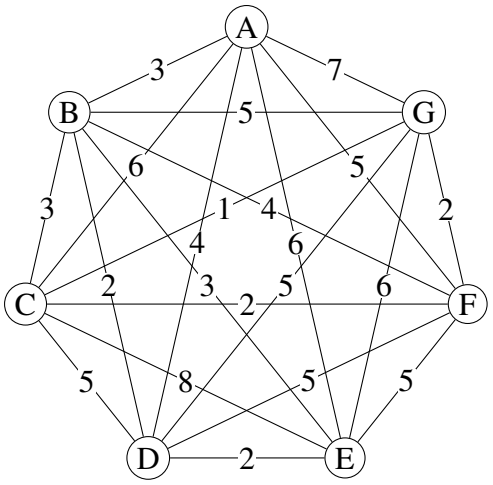
   A

   B G

   C F

   D E

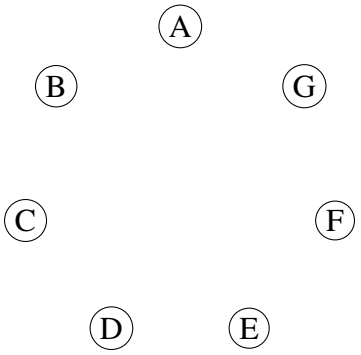
   circuit weight: _____.



edge	weight	edge	weight	edge	weight
CG	1	BE	3	DG	5
BD	2	AD	4	EF	5
CF	2	BF	4	AC	6
DE	2	AF	5	AE	6
FG	2	BG	5	EG	6
AB	3	CD	5	AG	7
BC	3	DF	5	CE	8

2. (4 points) Use the Cheapest Link Algorithm to find a Hamiltonian circuit in the graph (drawn above). Draw the circuit on the vertices below and determine weight of the circuit.

   Cheapest Link Algorithm: Select the cheapest edge in the graph that does not create a vertex of degree 3 or close the circuit too soon.



   circuit weight: _____.

3. (1 point) How many different Hamiltonian circuits does a graph on 7 vertices have? (It is sufficient to indicate how it is computed.)

4. (1 point) Is either algorithm, NNA or the Cheapest Link, **guaranteed** to give the cheapest Hamiltonian circuit in **every** graph? Explain your answer.